SGDAgent Reasoning Loop

Question: What is the maximum number of Taiyaki can the machine cook at the same time?
(A) 2. (B) 3. (C) 4. (D) 6.

Agent (Step 1):
Calling tool **Global-Level Retrieval** with query:
"Taiyaki machine cook at the same time".

Global-Level Retrieval:
Matched Events:
evt_0038 [29:10-29:40] score=2.6499, confidence=0.5736
batter, waffle_iron, machine; support: 1 segment(s)

evt_0040 [30:00-30:30] score=2.6286
taiyaki_mold, batter, taiyaki

evt_0041 [30:25-30:55] score=2.6386, confidence=0.7781
person, taiyaki, fish-shaped pastry; actions: holding
.....

The process of making Taiyaki

Agent (Step 2):
Calling tool **Segment-Level Retrieval** with query:
"taiyaki machine number of taiyaki can cook at the same time".

Segment-Level Retrieval :
Matched Segments:
seg_0073 [30:00-30:30] score=2.5423
entities: taiyaki_mold, batter, taiyaki, taiyaki_maker, pouring_tool; predicates: none

seg_0071 [29:10-29:40] score=2.4755
entities: batter, waffle_iron, machine, person

A Taiyaki machine with molds and fish-shaped pastry

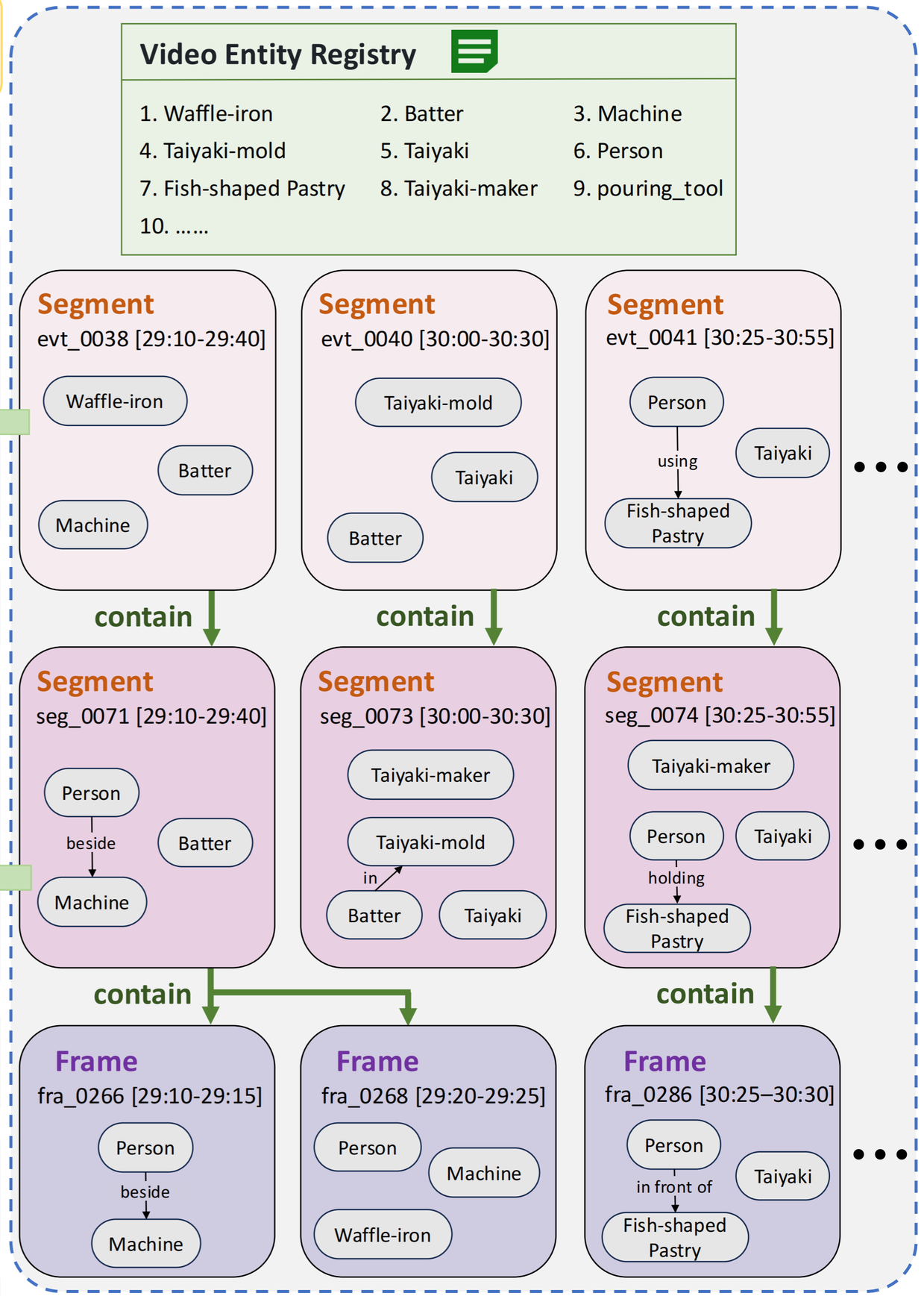
seg_0074 [30:25-30:55] score=2.5401
entities: person, taiyaki, fish-shaped pastry, taiyaki_maker; predicates: holding
.....

Agent (Step 3):
Calling tool **Frame Inspect** at ["00:30:00", "00:30:30"]:
"How many taiyaki can the machine cook at the same time?".

Frame Inspect:
Fine-grained Response:
The machine can cook 6 taiyaki at the same time.
The video frames **show 6 fish shapes** cooking simultaneously.

Miscounting molds versus fish-shaped pastries

Agent (Step 4):
Executing **Answer**: (D) 6.



Hierarchical Spatio-Temporal Scene Graph Representation



Long Video Input